

BTLO Challenge Report

Bruteforce — Incident Response

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Platform	Blue Team Labs Online (BTLO)
Category	Incident Response
Difficulty	Medium
Points	20
Tools Used	Kali Linux, grep, awk, WHOIS

1. Scenario

A system administrator identified a large number of Audit Failure events in the Windows Security Event log, consistent with an attempted RDP brute-force attack. The objective is to analyze the exported event log data to characterize the attack: how many attempts were made, which account was targeted, and where the traffic originated from.

2. Methodology

2.1 File Acquisition

The challenge archive was extracted with 7-Zip using the password provided on the challenge page (BTLO), producing four files: a CSV export, the raw .evtx event log, a duplicate .txt export, and a README. The CSV export was used for analysis since it is directly greppable, unlike the binary .evtx format.

2.2 Counting Audit Failure Events

```
grep -c "^Audit Failure," BTLO_Bruteforce_Challenge.csv
```

Result: 3,103 total Audit Failure events.

2.3 Identifying the Targeted Account

```
grep -A 3 "Account For Which Logon Failed" BTLO_Bruteforce_Challenge.csv | head -20  
grep "Account Name:" BTLO_Bruteforce_Challenge.csv | sort | uniq -c
```

The account name administrator appeared exactly 3,103 times — matching the total failure count exactly and confirming every single failed logon attempt targeted this one local account.

2.4 Failure Reason and Event ID

```
grep "Failure Reason:" BTLO_Bruteforce_Challenge.csv | sort | uniq -c
```

All 3,103 events returned the same failure reason: *"Unknown user name or bad password."* The Windows Event ID for these logon failures, visible directly in the CSV header, is 4625 — the standard event ID for a failed logon.

2.5 Source IP and Attribution

```
grep "Source Network Address:" BTLO_Bruteforce_Challenge.csv | sort | uniq -c
```

All attack traffic originated from a single IP address: 113.161.192.227. A WHOIS lookup was performed to attribute the IP to a country of origin:

```
whois 113.161.192.227 | grep -i country
```

The lookup returned a country code of VN, corresponding to Vietnam.

2.6 Source Port Range

```
grep -oP "Source Port:\s*\K[0-9]+" BTLO_Bruteforce_Challenge.csv | sort -n | sed -n '1p;$p'
```

Extracting and sorting all Source Port values numerically, then isolating the first and last lines, gave a port range of 49162–65534.

3. Findings

#	Question	Answer
1	How many Audit Failure events are there?	3103
2	Username of the targeted local account?	administrator
3	Failure reason related to the Audit Failure logs?	Unknown user name or bad password.
4	Windows Event ID associated with these logon failures?	4625
5	Source IP conducting this attack?	113.161.192.227
6	Country associated with this IP address?	Vietnam
7	Range of source ports used by the attacker?	49162-65534

4. Lessons Learned

- A CSV export of Windows Event Logs is far faster to triage with standard command-line tools (grep/awk) than parsing raw .evtx files directly, especially for high-volume, repetitive events like brute-force attempts.
- Cross-checking counts (e.g. confirming the targeted-account count matches the total failure count) is a simple but effective way to validate findings during an investigation.
- This exercise complements detection work already built for SMB brute-force (T1110) in a home Splunk lab — together they cover both the detection-engineering and investigation sides of a brute-force incident.